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Unveiling the energy price tag - Assessing the burden of household energy expenditures among European countries☆

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ABSTRACT

The equity dimension of climate and energy policy is increasingly emphasised both in the academic literature as well as by policymakers. We contribute to the growing body of research that scrutinizes these aspects by quantitatively assessing the patterns of in-come shares of energy expenditures using household survey data from European Union countries for the years 2010, 2015 and 2020. We employ established indices from the public finance literature to quantify the variation in energy expenditure shares along the income distribution. Furthermore, our analysis highlights that an assessment of the equity implications of policies needs a more nuanced approach than just considering energy poverty measures, since these policies affect households along the income distribution. Additionally, we investigate the role of various socio-demographic factors in shaping energy expenditure inequality by a decomposition analysis. In contrast to income, that leads to a higher concentration of energy expenditures, certain characteristics-such as household size and socio-economic status—contribute to a more equal distribution of energy expenditures in the population. Thus, a more differentiated and fine-tuned policy mix beyond purely income-based transfers may be required.

1. Introduction

In the current landscape, addressing climate change while simultaneously catering to the security of energy supply stand out as priorities for policymakers globally. The equity di-mension of soaring energy prices takes center stage in the realms of climate change research and the ongoing transition of modern energy systems towards renewable sources. Particu-larly noteworthy is the intersection of policies aimed at mitigating negative environmental impacts, such as carbon pricing, with the potential to escalate the costs of energy sources. This dynamic triggers unintended consequences on individual welfare, casting a shadow on the acceptability of such policies.

These developments impact households differently, contingent on their income and other socio-demographic characteristics. Numerous studies on climate change mitigation and the shift to renewable energy underscore the equity concerns of such policies (Hassett et al. (2009) and Nelson and Dodd (2023)). However, we still need to better understand the distribution of the energy expenditure burden as it is an unintended side effect of policies aiming at reducing carbon emissions. The equity implications have also been acknowledged by policymakers who, in the design of new policy packages such as the EU Fit For 55 package, underscore the importance of a socially fair transition.

In this paper we provide a detailed analysis of the distribution of the household energy expenditure-to-income ratio. Whereas previous studies emphasised the burden faced by the poorest households and hence the topic of energy poverty, we assert that energy expenditures affect households *along* the entire income distribution. We calculate several inequality indices based on data from the European Household Budget Survey (EHBS). These indices capture different dimensions of distributional outcomes and are relevant for informing targeted policy measures. The EHBS offers a wealth of information on energy-related housing expenditures, income, and socio-demographic characteristics for the years 2010, 2015, and 2020. The repeated cross-sectional data makes it possible to capture the development over the last decade,

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during which the climate policy mix has evolved.

We introduce new measures or customize existing measures from other fields to our literature strand. To start with, we calculate the average shares of income households spend on energy by income quintile. In a second step, we assess the rate at which these shares decrease along the income distribution. This shows whether the burden decreases overproportionally from the lowest to the highest income households, compared to a linear decline in shares. Third, we construct energy expenditure concentration curves — akin to income Lorenz curves — that illustrate the cumulative proportion of energy expenditures against the cumulative proportion of the sample. We then proceed to calculate so-called Kakwani indices that capture the magnitude of the negative income gradient of energy expenditure shares by comparing the relative concentration of energy expenditures with that of disposable income.[1] In addition, we decompose the energy expenditure concentration indices aiming to discern the specific contributions of household socio-demographic characteristics to the overall inequality in energy spending.

Our findings unveil a pattern of declining energy burdens with higher incomes across all analyzed countries, albeit with significant heterogeneities. First, there are significant disparities in the portions of income allocated to energy expenses not only across countries but also among different income brackets. The most notable gap in expenditure proportions between the top and bottom income quintiles occurred in Croatia in 2020, with a substantial difference of 19 percentage points, while Luxembourg exhibited the smallest difference at 4.7 percentage points. The descriptive evidence suggests that these large discrepancies can be partly explained by the types of heating systems that are more prevalent among poorer households (i.e., coal) since these are often also associated with higher costs. This implies that public policies aimed at supporting low-income households can be specifically targeted for instance, at facilitating the replacement of fossil-based heating systems with more environ-mentally friendly heating systems which are also cheaper in terms of their variable recurring fuel costs.

When considering the rate at which the share of energy expenditures decreases with higher income, for instance in 2020, countries such as Greece or the Netherlands stand out with the highest values in absolute terms, implying that the burden decreases overproportionally from the lowest to the highest incomes. In contrast, Croatia and Bulgaria even feature positive but small values indicating a lower degree of convexity or a proportional decrease of energy expenditure shares along the income distributions. In general, this degree of convexity of energy expenditure shares can help us determine whether only the specific segment of low-income households is disproportionately affected or, if the impact extends to a broader group including the middle class. Policy instruments can then be tailored based on these insights. Significant variations in the magnitude of the negative income gradient of energy expenditure shares as measured by the Kakwani index are also identified among the countries under scrutiny.

Furthermore, we conduct a decomposition of the concentration index, to discern the contributions of specific household socio-demographic characteristics to the overall inequality in energy spending. Equivalent income is the main driver of energy expenditure inequality, though its explanatory power varies— 50 % in Bulgaria to for instance, 83 % in Spain. Surprisingly, factors such as household size, age of the household head, and socio-economic status often contribute to a more equal distribution. Hence, a more differentiated policy mix beyond purely income based transfers may be appropriate to address these issues.

[1] The Kakwani index has been employed in the public economics literature to measure the progressivity of tax systems.

2. Literature

The empirical literature on the distributional impact of climate policies, energy poverty, and the energy transition provides valuable insights into the regressive nature of such measures, particularly in the context of the United States. Studies such as Burtraw et al. (2009), Hassett et al. (2009), and Fullerton et al. (2011) present evidence highlighting the regressive effects of US climate policy or underscore the importance of accounting for the equity effects of climate policies (Konisky and Carley (2021)). Examining electricity prices, Borenstein (2012) demonstrates that utilities adopt non-marginal increasing block prices and inefficiently low access fees to protect low-income customers while Levinson and Silva (2022) find that energy utilities concerned about inequality charge lower-than-efficient fixed costs and higher-than-efficient volumetric charges.

The contribution by Vona (2023) offers an overview of the literature on the distributional effects of environmental policies, including those related to energy. Overall, these papers consistently identify regressive effects on income.

Energy poverty, closely linked to energy inequality, has also been scrutinized by various researchers, including Drescher and Janzen (2021), Wu et al. (2017). For example, Drescher and Janzen (2021) use German SOEP data to reveal that, in 2019, 17 % of households spent more than 10 % of their income on domestic energy, with 4.5 %–14 % experiencing persistent energy poverty. Wu et al. (2017) demonstrate a nearly 50 % increase in energy poverty in the United States from 1990 to 2015.

We enhance the existing literature by providing a comprehensive overview of energy expenditures across several European countries. In doing so, we aim to draw a more nuanced picture that highlights the multiple dimensions necessary to assess inequality in this context. Looking at the literature using inequality indices in energy related research, we find that previous studies commonly employ Gini coefficients to illustrate energy expenditure inequality among households, with Jacobson et al. (2005) and Oswald et al. (2020) providing examples for various countries. However, we argue that the Kakwani index is better suited for analyzing energy expenditure inequality, particularly in the context of advanced economies. This is the case, since computing a Gini type index, just offers information on the cumulative share of energy expenditures across the population, which is meaningful in the analysis of energy access, however it does not relate this to the distribution of income. The Kakwani index though, captures this latter aspect as well. The index, initially designed to assess the progressivity of tax payments, has found applications in health economics to analyze the distribution of health expenditures, as evidenced by studies such as van Doorslaer and Wagstaff (1994), Paci et al. (1989), Kakwani et al. (1997), and van Doorslaer et al. (2003). Subsequently, the literature on the inequality of CO_2 emissions adopted the Kakwani index to measure and compare the regressivity or progressivity of emission distributions across countries/regions and over time. (Clarke-Sather et al. (2011); Cantore and Padilla (2010); Padilla and Serrano (2006)). Both the health economics and climate economics literature rely on concepts of regressivity and inequality originating from public economics, even though these papers do not explicitly analyze taxes or policies. Instead, these concepts are applied to assess the distributional effects of expenditures or emissions. Huang (2022) employs the Kakwani index to illustrate electricity expenditure inequality in Taiwan, marking a notable exception in the literature's use of this index for assessing energy expenditure inequality. Recent papers by Borenstein and Davis (2024) and Borenstein and Davis (2016), though not explicitly calculating the Kakwani index, assess the distributional effects of energy tax credits and find that the bottom three income quintiles have received about 10 % of all credits, while the top quintile has received about 60 %. The authors plot income concentration curves against concentration curves for federal income tax credits, however they do not compute any indices of regressivity and focus on tax benefits in the United States while we

consider the distribution of overall household energy expenditures in different European Union countries. The concept of regressivity has been usually applied in the context of taxes (or tax credits or other policy instruments). Taxes are generally perceived as burdensome, as they reduce the disposable income available to cover other needs. In this sense, energy expenditures are only partly comparable to taxes. Nevertheless, certain elements of energy consumption — such as heating or cooling to maintain a reasonable indoor temperature and energy for cooking—can be considered as basic needs. Hence, the expenditures spent on these energy services may be quite analogous to tax expenditures due to the non-discretionary nature. This line of reasoning mirrors approaches rooted in health economics to assess the regressivity of out-of-pocket payments (Rasell and Bernstein (1995); Wagstaff et al. (2020)). In contrast, expenditures on non-essential energy uses, such as operating a sauna, are clearly of a different nature, turning this to a more complex issue than in the case of a tax.

Our study thus contributes to this growing body of literature by augmenting the dimensions according to which energy expenditure inequality should be assessed.

3. Data

We utilize data from the European Household Budget Survey (EHBS), provided by Eurostat, containing comprehensive details about individuals in households, household income, and expenditures. The surveys in the majority of countries except for Germany (which employs quota sampling) are based on probability sampling. This means that every unit (i.e., private households) in the study population is given a known, non-zero chance of selection into the sample.[2] Our analysis focuses on sample surveys conducted in 2010, 2015, and 2020, encompassing between 151,687 and 254,942 observations.[3] The number of observations by country and year lies between 1405 (Luxembourg in 2020) and 53,820 (Germany in 2010).[4] To ensure comparability across households, we compute equivalised values for the disposable income.[5] Disposable income is defined as gross income minus expenditures on mandatory social security, taxes, basic health insurance premiums, and regular transfer payments to other households. The EHBS dataset also provides information on housing-related energy expenditures, encompassing electricity, gas, district heating, and fuel categories.[6]

4. Methods & results

In this section, we assess several dimensions of the magnitude of the distribution of the energy expenditure-to-income ratio. First, we compare the share of income spent on energy between the lowest and highest income groups. Second, we examine the rate at which this share declines across the income distribution. Third, we evaluate the income gradient of energy expenditure shares by comparing the distributions of energy expenditures and income. To this end, we apply the Concentration and Kakwani indices, which are further decomposed by key socio-demographic variables in the final stage of the analysis.

4.1. Energy expenditure as a share of income

The most significant contrast in expenditure shares between the highest and lowest income quintiles is observed in Croatia in 2020, with a substantial 19 percentage points difference, while Luxembourg exhibits the smallest difference at 4.7 percentage points (see Fig. 1 below).[7] This difference in the burden between the first and fifth income quintile is com-parable to the average tax burden for different income brackets in the context of measuring tax progressivity. However, this is only a first approximation and requires a more nuanced scrutinization as explained further below.

One possible explanation for these marked differences, can be the influence of heating type. In columns 2–6 of Table A1 in the Appendix we present the differences in the shares of various heating systems between the first income quintile (Q1) and the fifth income quintile (Q5). For instance, in Croatia, the country with the largest difference in the share of energy expenditure burden between the richest and the poorest households, the share of coal heating is almost 5 percentage points higher for Q1 than for Q5 in 2020, whereas in Belgium, one of the countries with the lowest difference in the burden, heating systems of all types are evenly distributed among income groups. Furthermore, we compute the average cost associated with each heating type for each country in 2020. This is presented in columns 7–11 of Table A1. Thus, Croatian households using coal heating face an average expenditure share of 23.4 %, which is the highest among all heating types. Hence, this descriptive evidence suggests that low-income households are more likely to employ heating systems that command higher expenditure shares.

4.2. Assessing the degree of convexity

One intuitive measure of the magnitude of the decline in energy shares along income—outlined above—is the difference in the share of income spent on energy between the lowest and high-est income households. A second, often overlooked dimension, is the rate at which this share declines along the income distribution. A disproportionately steep decline implies a higher steeper negative income gradient than a

[2] Since the survey is conducted based on a gentleman's agreement, each EU country decides the objectives, methodology, and frequency of the survey. Data collection involves a combination of one or more interviews and diaries maintained by households and/or individuals, generally on a daily basis. For more information regarding the survey and sampling process see European Commis-sion, European Commission (2003): https://ec.europa.eu/eurostat/documents/54431/1966394/Methodology+and+recommendationsl2003EN.pdf/51442ee8-ec72-0ded-43ab-54089f0ef628?t=1694098227856.

[3] From this extensive pool, 354,031 observations were meticulously selected for analysis. The exclusion criteria were primarily based on the presence of missing values in crucial categories, such as income and energy expenditures, as well as the geographical context of the respective country. Furthermore, to ensure the reliability and representativeness of our findings, we systematically eliminate observations that did not fall within the range of a household's share of income spent on energy between 0 and 1.

[4] In 2015, the number of observations per country was as follows: Germany 52,248, Poland 36,534, Romania 30,443, Spain 21,906, France 16,635, Netherlands 14,358, Portugal 11,294, Hungary 7,131, Austria 7,045, Ireland 6,797, Belgium 6,128, Greece 5,980, Slovakia 4,780, Latvia 3,833, Slovenia 3,727, Malta 3,683, Estonia 3,387, Lithuania 3,435, Luxembourg 3,165, Bulgaria 2,959, Czech Republic 2,924, Cyprus 2,870, Sweden 2,855, Denmark 2,193, Croatia 2013.

[5] Equivalised income is obtained by dividing the disposable income by the number of equivalent adults in the household, adhering to the OECD equivalence scale. This scale assigns a weight of 1 for the first adult, 0.5 for each additional household member aged 14 and above, and 0.3 for each additional household member under 14.

[6] We follow the same methodology to compute equivalized energy expenditures.

[7] As a robustness check, we calculate the same shares using only households with a share of income spent on energy below 50 % (see Figure A1 in the Appendix). Overall, the ranking of the countries does not change substantially: most countries shift by no more than one rank or keep the position. Notable exceptions include France and Bulgaria, which move up by five and three ranks respectively, toward the group of less regressive countries. In contrast, Austria moves down by three positions. Furthermore, for most countries, the difference in energy expenditure shares of the first and last income quintile changes only slightly compared to Fig. 1. The largest change is observed for Croatia, where the difference decreases to around 14 percentage points.

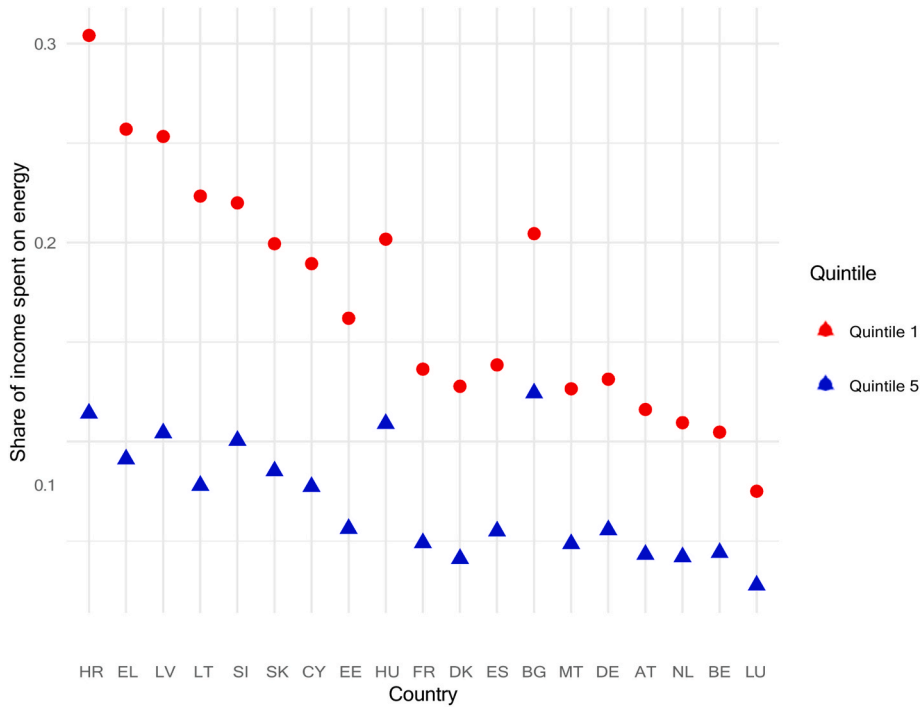

Fig. 1. Shares of total energy expenditures in equivalent income for the first and fifth income quintile in 2020.
Note: The figure depicts the shares of total energy expenditures in equivalent income for the first and fifth income quintile, ranked by the difference between $Q1 - Q5$.
Source: Own illustration based on data from Eurostat.

linear decrease would. The intuition is comparable to the analysis of tax progressivity, where a progressive (regressive) tax system is characterized by average tax rates that increase (decrease) with income. Furthermore, if the rate at which the average tax rate increases (decreases) is increasing (decreasing), the tax structure is even more progressive (regressive).

This relationship is also closely linked to the concept of Engel curves, which describe how household consumption of specific goods varies with income. The shape of the Engel curve depends on whether the good is a necessity or a luxury: it is typically concave downwards for necessities and convex downwards for luxuries.[8] For energy-related goods, differences in income elasticities can explain the distribution of energy expenditures across the income spectrum. Krishnamurthy and Kriström (2015) estimate a low residential electricity income elasticity of less than 0.16 for a sample of 11 OECD countries whereas Espey (1998) employs a meta analysis and finds a median short run income elasticity for gasoline of 0.39 and a more recent study by Havranek and Kokes (2015) an even lower value of 0.1. This suggests that these goods are necessities, as they require a minimum level of consumption regardless of income — rendering them comparable to compulsory expenditures such as taxes. As a consequence, the budget share spent on energy is expected to decline with income, albeit at a decreasing rate. While several necessities, such as food, exhibit income elasticities below unity and hence declining budget shares with rising income, energy consumption warrants special attention. Unlike food, where households can often adjust quality or composition and where many countries exempt basic items from taxation (or levy lower taxes), energy services are essential and largely non-substitutable (at least in the short run). Low-income households are often locked into inefficient housing or appliances, so they face higher energy intensity per unit of service, and because energy is a direct target of carbon pricing and environmental taxation, its distributional consequences are particularly salient compared to other necessities with similar income elasticities.

While the concept of Engel curves describes the shape of the relationship between consumption of specific goods and income, we also quantify this by creating an index that facilitates comparison between countries and over time.

Hence, we compute a measure that captures the degree of convexity of energy expenditures, by assessing the rate at which average energy expenditure shares decrease with higher income.

First, we rank households by income within each country-year and group them into B income bins of equal size (e.g., $B = 500$). Let $b \in \{1, 2, \ldots, B\}$ denote the income group index ordered from the lowest to the highest income group. Let $\bar{e}_{bjt}$ be the observed average energy expenditure share in group b in country j in year t, and let $\hat{e}_{bjt} = a_{jt} - d_{jt} \cdot b$ represent a hypothetical linear decline in the share from the lowest to the highest income group, where:

$$d_{jt} = \frac{\overline{e_{1jt}} - \overline{e_{500jt}}}{B - 1} \tag{1}$$

We then define the *degree of convexity (DOC)* as:

$$\mathrm{DOC}_{jt} = \frac{1}{B} \sum_{b=1}^{B} \left(\overline{e_{bjt}} - \widehat{e_{bjt}} \right), \tag{2}$$

which captures the average deviation of the actual distribution from the assumed linear trend.

The degree of convexity is however best understood by looking at Fig. 2. It illustrates positive (black) and negative (grey) deviations between the actual observed energy expenditure burdens and a linear decreasing trend. The degree of convexity is then obtained by aggregating these positive and negative deviations over the income of households observed in each survey. As a specific example, Fig. 3 depicts the shares of income spent on energy by different income groups ordered

[8] Note that, in contrast to classic Engel curves, which plot absolute consumption levels against income, we plot the share of income spent on energy. This transformation implies that convex shapes indicate normal goods, as budget shares decrease with rising income.

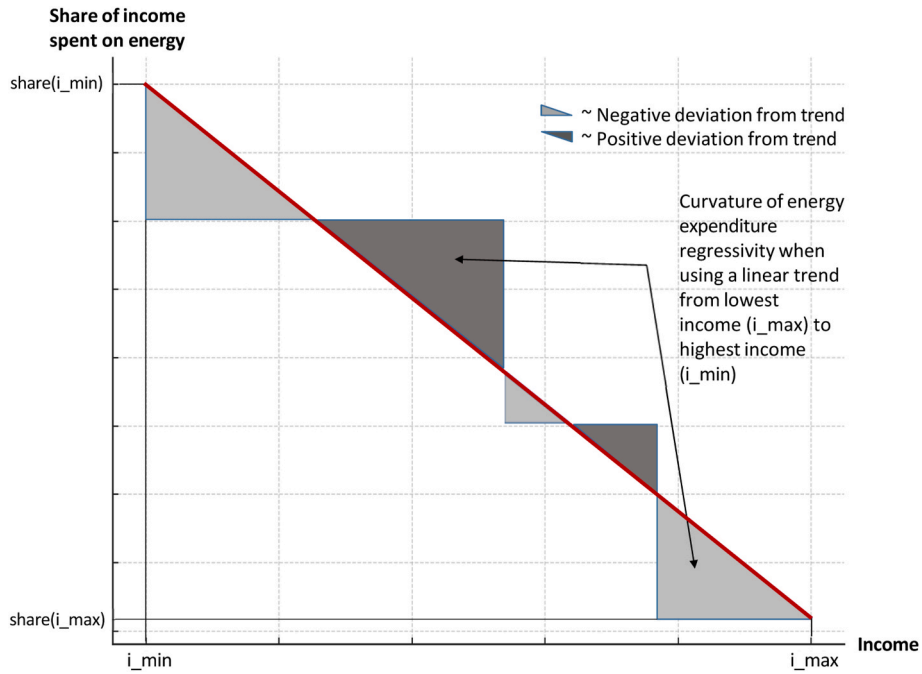

Fig. 2. Curvature of energy expenditures.
Notes: Red line assumes energy expenditure shares decrease linearly with income.
Source: Own illustration.

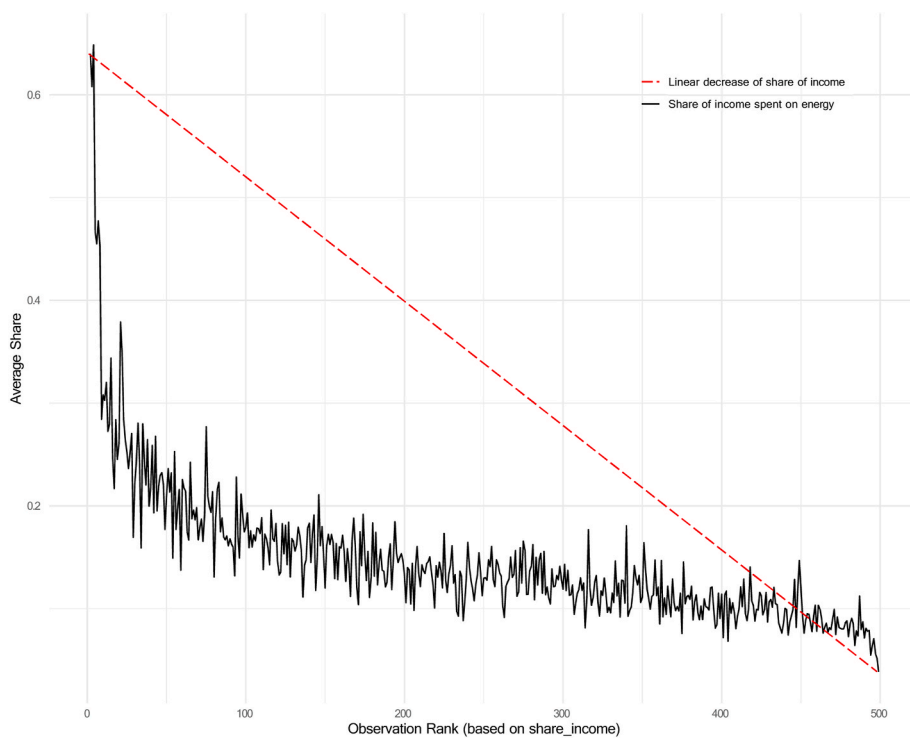

Fig. 3. Actual vs. linearly decreasing shares of energy expenditures in income for Greece in 2020.
Notes: Red dashed line assumes energy expenditure shares decrease linearly with income. This is an illustrative example. Additional graphs for other countries and years can be provided upon request.
Source: Own illustration based on data from Eurostat.

from lowest to highest income for Greece and the year 2020.[9] After ranking households within a given country and year by income, we group observations into 500 bins of equal size.[10] Then, we calculate the average share of income spent on energy within each bin and its deviation to the linear trend. Lastly, we derive the average deviation. The red dashed line shows a hypothetical distribution of linearly declining shares.[11] The observed shares lie in most cases substantially below the dashed line. Thus, in reality, the energy expenditure rate decreases overproportionally.

More negative values of the index thus represent a higher degree of convexity and an overproportionally decreasing burden of energy expenditures with higher income. Table 1 presents the results for all analyzed European countries.

Compared to the analysis in subsection 4.1, the DOC captures the magnitude of the negative gradient of energy expenditure shares in a distinct way. For instance, Croatia and Bulgaria exhibit positive but small values for the DOC, suggesting a roughly proportional decline in the burden of energy expenditures across the income distribution. This stands in contrast to the previous analysis, in which these countries appeared among the more extreme cases. Moreover, Greece stands out in both 2020 and 2015 with the highest DOC, followed by the Netherlands. At first glance, this may appear surprising for the Netherlands, as Fig. 1 indicates only a modest difference in average expenditure shares between the lowest and highest income quintiles. This highlights that just looking at the difference between average expenditure shares among some aggregated income groups does not provide an unambiguous assessment of the distribution of the energy expenditure burden.

These discrepancies can be attributed to two key factors. First, the DOC is partially driven by the difference in average expenditure shares between the lowest and highest income *bins*—specifically, the 1st and 500th bin—rather than between the first and fifth quintile. Since this approach relies on the extremes of the income distribution, the observed difference tends to be larger than the corresponding quintile-based measure. To illustrate the first driver of the differences between the two measures, Figure A2 displays the average energy expenditure shares for the 1st and 500th income bin. In contrast to Fig. 1—countries such as the Netherlands and Luxembourg exhibit substantial differences in these bin-based expenditure shares, comparable to those observed for Greece.

Table 1
Degree of convexity.

Country	2010	2015	2020
AT			−0.058
BE	−0.179	−0.120	−0.083
BG	−0.098	−0.108	0.002
CY	−0.125	−0.023	−0.023
CZ	−0.103	0.062	
DE	−0.076	−0.062	−0.122
DK	−0.126	−0.061	−0.025
EE	−0.136	−0.188	−0.044
EL	−0.179	−0.229	−0.232
ES	−0.098	−0.073	−0.057
FI	−0.081	−0.114	
FR	−0.125	−0.143	−0.143
HR	−0.099	0.008	0.053
HU	−0.080	−0.179	−0.162
IE	−0.201	−0.165	
LT	−0.117	−0.094	−0.044
LU		−0.131	−0.202
LV	−0.099	−0.075	−0.101
MT	−0.111	−0.131	−0.131
NL		−0.217	−0.210
PL	−0.098	−0.043	
PT	−0.074	−0.147	
RO	0.003	−0.001	
SE	0.038	0.037	
SI	−0.183	−0.110	−0.127
SK	0.108	−0.061	−0.113
UK	−0.094		

Note: This table presents the degree of convexity of energy expenditure shares along the income distribution following the calculations described in section 4.2. After ranking households within a given country and year by income, we group observations into 500 bins of equal size.

This helps explain part of the discrepancy between the rankings derived from the DOC and those based on quintile-level comparisons.

Second, the average (negative) deviation from a linear trend across the income distribution increases when extremely high expenditure shares are concentrated at the lower end of the income distribution and/or when there are no comparably low values at the upper end. These distributional features can amplify the measured convexity, even in cases where the quintile-based measure suggests a smaller disparity. Thus, the DOC reflects not only the absolute difference in expenditure shares between the lowest and highest incomes, but also the steepness of the decline in these shares along the income distribution.

4.3. Concentration and Kakwani index

In this section we apply metrics originating in public economics to assess the progressivity of tax systems. There is a lack of consensus on how to measure this. However, the literature distinguishes between approaches that are based on tax liability progression and approaches based on the distribution of tax liability. The former employs the elasticity of tax liability at a given income level and exploits information on average versus marginal tax rates (Musgrave and Thin, 1948). Thus, it can only provide a picture of progressivity at a given point in the income distribution. The latter approach measures tax progressivity in terms of tax liability shares relative to income shares across the income distribution. Thus, it accounts for both the income distribution as well as the income tax schedule in one measure. The two measures commonly applied are the so-called Kakwani index (Kakwani, 1977) and the Suits index (Suits, 1977). Both measure the extent to which the tax system deviates from proportionality. They differ from the former approaches also insofar they yield a summary statistic whereas the former varies with the tax base. If the researcher is interested in the impact of taxes on the income distribution, the Musgrave and Thin (1948) approach is adequate since it relates the before and after Gini coefficients. In contrast, if we are interested in how the percentage distribution of taxes compares to the percentage distribution of income, the latter approaches

[9] We choose for illustration purposes Greece since it features the highest absolute value for the degree of convexity in 2020. See Table 1.

[10] We replicate the analysis to assess the robustness of our results using 100 bins instead of 500, each including an equal number of observations. The results of this alternative computation are presented in table Table A2 in the Appendix. For some countries the relative ranking continues to hold whereas for others, the trend and the sign are reversed. For instance, Croatia and Romania exhibit a reversal in both the sign and the direction of the trend while for Sweden only the signs change. The following countries also experience changes in the trend of the DOC in at least one year: Bulgaria, Germany, Denmark, Hungary, Lithuania, the Netherlands, and Slovenia. While some changes are only minor — such as in the Netherlands — or the overall trend between 2010 and 2020 is maintained — as in Bulgaria — others are more substantial, as in Denmark or Lithuania. The general trend can be explained in the following way: with higher bin size (as the number of bins is reduced, each bin includes a higher number of observations), between-group variation is smoothed out, leading to less pronounced deviations from the linear trend. This reflects the following trade-off — on the one hand, a larger number of bins provides a more detailed view of the income distribution, on the other hand it increases sensitivity to outliers. The aforementioned comparison in average energy expenditure shares between the lowest and highest income quintiles already provides a more aggregated perspective. Nevertheless, a more differentiated approach using a higher number of bins remains valuable for capturing nuanced patterns across the income distribution.

[11] Figure A3 replicates Fig. 3, excluding bins 1 and 500. While the slope of the red line is slightly reduced since the shares now vary between 64 % and 3.8 %, instead of 73 % and 2.9 %, the overall pattern remains largely unchanged.

should be applied.

In this paper, we are interested in the distribution of the burden of household energy expenditures across income quintiles as opposed to how these may themselves affect the income distribution. Hence, the Kakwani index is the suitable approach. Developed by Kakwani (1977) as a measure of tax progression, the index is originally defined as the difference be-tween the concentration index of tax payments and the Gini index (see Kakwani (1977)). In a similar manner, we can apply it to energy expenditures. Hence, the Kakwani index for energy expenditures in country j and year t is defined as:

$$K_{jt} = C_{jt} - G_{jt} \tag{3}$$

where C_{jt} and G_{jt} denote the energy concentration and Gini index respectively. The energy concentration index for each country $j = 1, \ldots, J$ and year $t = 2010, 2015, 2020$, is defined as

$$C_{jt} = \frac{2}{n \cdot \mu_{jt}} \sum_{i=1}^{n} exp_{ijt} R_{ijt} - 1 \tag{4}$$

where exp_{ijt} denotes the specific type of energy expenditure (i.e., electricity, heating, mobility) in the respective country j in year t, $\mu_{jt} = \frac{1}{n}\sum_{i=1}^{n} exp_{ijt}$ the mean energy expenditures in the respective country and year, and R_{ijt} the relative rank of the i-th household in the income distribution of the respective economy. C_{jt} ranges between -1 and 1 and is defined as twice the area between the concentration curve and the diagonal.[12] The further the concentration curve lies from the diagonal, the more unequally energy expenditures are distributed across income groups. When the diagonal and the concentration curve overlap $C_{jt} = 0$. It is negative (positive) when the concentration curve lies above (below) the diagonal. If it is positive, it means that energy consumption expenditures are concentrated predominantly among high-income households. The concentration index is a measure of *relative* inequality, meaning that a doubling of everyone's energy expenditures leaves the index unchanged. The Gini index, is defined as:

$$G_{jt} = \frac{\sum_{i=1}^{n}\sum_{l=1}^{n} |x_{ijt} - x_l|}{2n\overline{x_{jt}}} \tag{5}$$

with x_{ijt} the disposable income of the ith household, and $\bar{x}_{jt}$ denoting the mean income in the respective year and country. By definition, $-2 \leq K_{jt} <= 1$. If the concentration curve of energy expenditures lies above the income Lorenz curve (as in Fig. 4), $K_{jt} < 0$ which means that the cumulative share of energy expenditure exceeds the cumulative share of income and thus implies declining energy expenditure ratios with higher income. The opposite is true if $K_{jt} > 0$. [13]

The following figure displays as an example the income Lorenz curve and the concentration curve for overall household energy expenditures for Bulgaria and the year 2020 since it displays the least negative Kakwani index in 2020.[14] On the x-axis households are ordered by income while on the y-axis we see the cumulative expenditure or income up to the corresponding income percentile on the y-axis. Fig. 4 illustrates that total energy expenditures are less concentrated than income in Bulgaria. The Lorenz and the energy expenditure concentration curve allow for a direct comparison of the cumulative share of income and energy expenditures along the income distribution. In contrast, comparing the income Lorenz curve to a Lorenz curve based on a ranking of households according to their energy expenditures would not be appropriate, as the underlying population rankings would differ. Such a comparison cannot be captured within a single graph, since the two curves would not refer to the same ordering of households.

The dataset enables us to distinguish between various categories of energy expenditures such as expenditures on mobility or housing. Our analysis consistently reveals negative values of the index across all countries studied, with a median value of the Kakwani index for total household energy expenditures of -0.18 in 2020. Fig. 5 depicts the median values of the Kakwani index for 2010 and 2020 across different types of energy expenditure categories. The graph illustrates that housing-related energy expenditures (including expenditures on electricity and heating) feature the most negative values of the index. As mentioned above, mobility expenditure ratios feature lower absolute values but probably simply as a result of the poorest households not owning a car and hence not incurring any fuel costs.[15]

Over the decade, only three countries—Belgium, Slovenia, and Hungary—experienced positive changes in the Kakwani index (Fig. 6), while in all others it remained stable or decreased. In Hungary and Slovenia, the change is solely due to rising concentration indices, as Gini coefficients remained largely unchanged. In contrast, in Belgium income inequality declined.

Between 2010 and 2020, concentration indices changed only marginally. Table A3 sum-marizes these changes for 16 EU countries. The most notable increase occurred in Hungary (+0.029), while Lithuania saw the largest decrease (-0.090). In 14 of 16 countries, the concen-tration index declined, suggesting more equally distributed energy expenditures. Meanwhile, income inequality remained largely stable. [16]

The preceding analysis demonstrates that the choice of metric significantly influences both the assessment of the magnitude of the decline in energy expenditure ratios along the income distribution and the ranking of countries. Hence, the simple comparison of energy expenditure income ratios across income classes is not sufficient. Table 2

[12] The 45-degree line (in black) is visible in Fig. 4 beginning in the origin.

[13] The concept of the concentration index is appealing due to its comparability with the Gini index, which allows for the derivation of the Kakwani index. In fact, the concentration index preserves the ranking of households from the Gini index, as it is based on each household's position within the income distribution. Therefore, the difference between the two indices compares the cumulative income and expenditure shares of the same households. In contrast, a Gini index for energy expenditures does not suffice, as household rankings by energy spending do not necessarily align with income rankings.

[14] The most negative Kakwani index is computed for Denmark. This country features the lowest value for the concentration index among the analyzed countries in 2020. This implies that energy expenditures in Denmark are more equally distributed along the income distribution compared to other countries. At the same time, the income inequality itself, as measured by the Gini index, is approximately equal to the median value of the analyzed countries. Hence, the difference between the two indices is highest for Denmark.

[15] One possible explanation for the evolution of Kakwani indices of fuel expenditures between 2010 and 2020 is the rising car ownership among lower-income households. Between 2017 and 2023, the proportion of German households without a car declined by two percentage points. This overall decrease was not mirrored across all economic strata: the share of households without a car remained stable within the "very low", "medium", "high", and "very high" income groups. In contrast, the share of households without a car fell by 12 percentage points within the "low" income group. (Mobilität in Deutschland (2023) and Mobilität in Deutschland (2017)). As these households acquire private vehicles, fuel costs account for a larger proportion of their income relative to the share spent by higher-income households, thereby amplifying the burden of fuel costs. Additionally, the growing adoption of electric vehicles — predominantly among wealthier households (Brückmann et al. (2021); Bigler and Radulescu (2025)) — may further reinforce this trend, as these households depend less on conventional fuels.

[16] Summary statistics for the concentration, Kakwani, and Gini indices for 2010, 2015, and 2020 are provided in Tables A4–A6 in the Appendix.

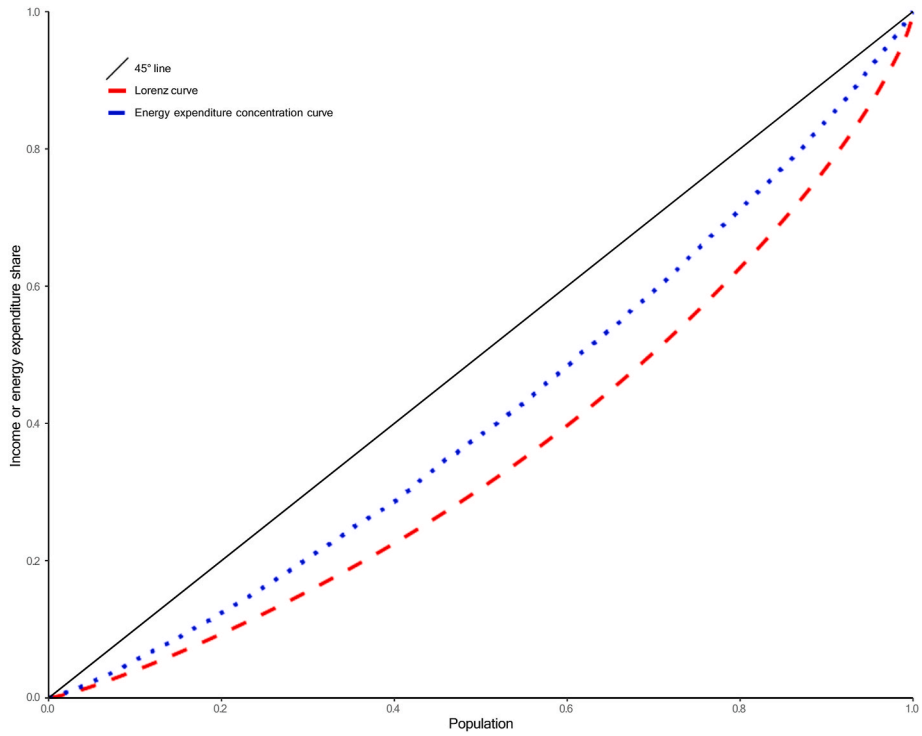

Fig. 4. Lorenz curve and concentration curve for energy expenditures in Bulgaria (2020.
Notes: This is an illustrative example. It shows the deviation of the distribution of income (in red) and energy consumption (in blue) from an equal distribution (in black). Additional graphs for other countries and years can be provided upon request.
Source: Own illustration based on data from Eurostat.

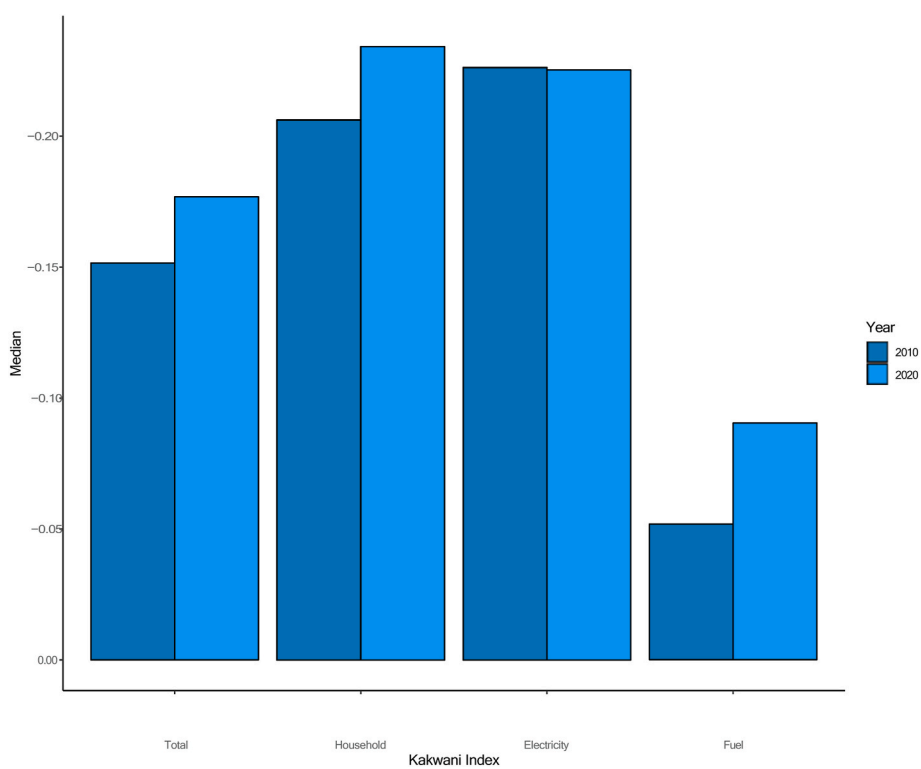

Fig. 5. Median Kakwani Index of European Union Countries in 2010 and 2020 for different energy expenditure categories.
Source: Own illustration based on data from Eurostat.

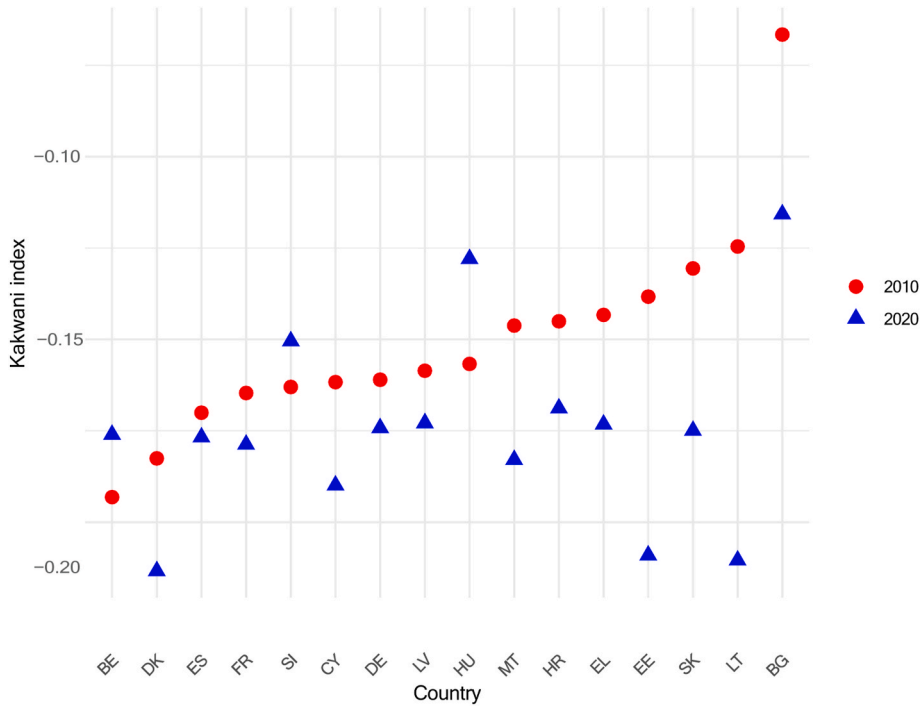

Fig. 6. Development of Kakwani indices in various European Union countries.
Source: Own illustration based on data from Eurostat. Austria, Luxembourg, and the Netherlands are not included in the graph, as data are missing for either 2010 or 2020, making a comparison infeasible.

Table 2
Energy expenditure inequality indicators for 2020.

Country	DOC	Gini Income	Concentration index	Kakwani index
AT	−0.058	0.272	0.079	−0.193
BE	−0.083	0.230	0.054	−0.176
BG	0.002	0.282	0.167	−0.116
CY	−0.023	0.331	0.142	−0.190
DE	−0.122	0.274	0.100	−0.174
DK	−0.025	0.265	0.052	−0.213
EE	−0.044	0.294	0.085	−0.209
EL	−0.232	0.289	0.116	−0.173
ES	−0.057	0.288	0.112	−0.177
FR	−0.143	0.346	0.167	−0.179
HR	0.053	0.299	0.130	−0.169
HU	−0.162	0.266	0.138	−0.128
LT	−0.044	0.301	0.091	−0.210
LU	−0.202	0.283	0.081	−0.203
LV	−0.101	0.330	0.157	−0.173
MT	−0.131	0.287	0.104	−0.183
NL	−0.210	0.265	0.088	−0.177
SI	−0.127	0.253	0.102	−0.150
SK	−0.113	0.220	0.045	−0.175

Source: Own computations based on data from Eurostat.

compares results across different indices for the countries analyzed.

For example, while Bulgaria and Cyprus exhibit similar Degrees of Concentration (DOCs), Bulgaria has the lowest (absolute) Kakwani index. In contrast, Cyprus shows a higher (abso-lute) Kakwani index. As illustrated in Figure A4, this discrepancy can be partially explained by the more equal distribution of energy expenditures across income groups in Cyprus, de-spite its more unequal income distribution (as reflected in a higher Gini coefficient compared to Bulgaria).

Interestingly, differences in the Kakwani indices for Hungary, Germany, Denmark, the Netherlands, and Austria are entirely attributable to variations in their concentration indices, since all five countries have similar Ginis in 2020. Furthermore, although the Netherlands and Austria display similar Kakwani indices, their DOCs differ substantially. As shown in Figure A5, this divergence is primarily driven by the lowest income percentiles in the Netherlands, which allocate a significantly higher share of their income to energy expenditures compared to the rest of the population.[17]

Additionally, country rankings vary depending on the metric used. When applying the DOC, Greece, the Netherlands, and Luxembourg emerge as countries with the steepest decline in the energy burden with higher income, while in Croatia, Bulgaria, and Cyprus the decline is less steep. However, using the Kakwani index yields a different ranking: Denmark, Lithuania, and Estonia feature the most negative income gradient of the energy expenditure-to-income ratio, whereas in Bulgaria, Hungary, and Slovenia rank the gradient is less steep. The use of multiple indicators provides a more nuanced understanding of the underlying causes of the burden in energy expenditures. These may stem from unequal income distribution, unequal distribution of energy expenditures, or disproportionately high energy burdens on the lowest income groups. Accordingly, policy responses can be tailored to address each of these drivers, depending on the specific context and societal preferences.

4.4. Socio-demograhic factors contribution to energy expenditure concentration

In the next step, we decompose the energy concentration index to quantify the contribution of distinct socio-demographic factors to energy expenditure inequality. Following van Doorslaer et al. (2003) and Huang (2022) we regress our variable of interest exp_{ijt} on a number of controls k:

$$exp_{ijt} = \alpha + \sum_k \beta_{kjt} x_{kijt} + \epsilon_{ijt} \tag{6}$$

β_{kjt} thereby represent the respective coefficient of control variable k and

[17] These differences are not due to aggregation or sample size variations; the survey sample includes 10,182 observations for the Netherlands and 7045 for Austria.

ϵ_{ijt} the error term. We use the following control variables: equivalent income, number of income earners, house-hold size, tenant status, and dummy indicators accounting for socio-economic status and household type.[18] Plugging eq. (6) into eq. (4), the concentration index C_{jt} can be expressed as [19]:

$$C_{jt} = \sum_k \frac{\beta_{kjt}\overline{x_{kjt}}}{\mu_{jt}} C_{kjt} + \frac{GC_{\epsilon jt}}{\mu_{jt}} \tag{7}$$

$$GC_{\epsilon jt} = \frac{2}{n}\sum_{i=1}^{n} \epsilon_{ijt} R_{ijt} \tag{8}$$

μ_{jt} thereby represent average energy expenditures in country j in year t, $\bar{x}_{kjt}$ the mean of the variable considered and C_{kjt} the concentration index of the respective variable in country j and year t. $GC_{\epsilon jt}$ is a generalized concentration index for the residual. Hence, the en-ergy concentration index can be decomposed into a deterministic component $\sum_k \frac{\beta_{kjt}\overline{x_{kjt}}}{\mu_{jt}} C_{kjt}$ defined as the weighted sum of the concentration indices of the regressors and a residual component representing the inequality in energy expenditures that cannot be explained by systematic variations in the regressors. The weight of the respective concentration indices C_{kjt} is given by the elasticity of energy expenditures with respect to the respective regressor $\frac{\beta_{kjt}\overline{x_{kjt}}}{\mu_{jt}}$.

We present results for two countries - Bulgaria and Spain and the year 2020. We choose these as illustrative examples, since among the countries for which we also have nonmissing observations for the included control variables, the two countries feature the lowest and high-est absolute values of the Kakwani index. Table 3 displays the results of an OLS regression. All coefficients should be interpreted with the reference category being the indicator variable with the lowest numerical value, except for equivalent income, which is a numeric variable measured at the level scale. In a next step, we compute the elasticities of energy expenditure with respect to the regressor of interest ($\frac{\beta_{kjt}\overline{x_{kjt}}}{\mu_{jt}}$).[20] The elasticity informs us how much the outcome depends on the respective regressor. For instance, if β_{kjt} is large, the outcome (i.e., energy expenditures) is sensitive with respect to changes in the respective control variable x_{kjt}. The largest values for the elasticities in absolute terms are recorded for equivalent income, the age of the household head and different household types. Columns (*CI*) report the values for the concentration index for each regressor (C_{kjt}) for the corresponding explanatory variables (i.e income, household size, etc. - similar to eq. (4) and as presented in eq. (B.5) in the Appendix). This measure illustrates how the respective control variable is distributed among different income groups, thereby indicating the level of inequality within an explanatory variable. This means that if a particular variable is more concentrated among higher income households and this variable also increases energy spending, then the inequality of energy expenditures will increase. The impact of each element on the disparity in energy expenses is ascertained by multiplying the concentration index with the elasticity of the respective determinant. The concentration indices are weighted by their respective elasticities and then aggregated by base variable. [21]

In Table A7 in the Appendix we present the fraction of inequality in energy expenditures explained by different components. Thus, in our illustrative example for Bulgaria and Spain, equivalent income explains only 50 % in Bulgaria but even up to 83 % of the energy expenditure concentration index in Spain. Covariates such as education, number of income earners, tenant status[22] or socio-economic status account for smaller fractions of the overall energy expenditure inequality. In contrast, some covariates such as the age of the household head (Bulgaria) or the household size (Spain) even contribute to a decrease in the concentration of energy expenditures.

The overall contributions of all explanatory variables to the concentration of energy expenditures are obtained by summing up the individual contributions. These results are presented for 15 countries for the year 2020 in Table 4.[23] The contribution of the deterministic component ranges between 0.03 (Belgium) to 0.18 in Bulgaria. On the whole, the incorporated controls account for 76 %–101 % of the overall concentration index.

For the same reason 4 countries - Malta, Slovenia, Luxembourg and the Netherlands - had to be excluded from the analysis.

Fig. 7 illustrates the relative contributions of different socio-economic variables to explaining the concentration index for various European Union states in 2020. Equivalent income stands out as the main contributor to the concentration index, albeit its contribution varies to a large extent between countries. Hence, in Latvia the contribution amounts to only 31 % whereas it is even as high as 97 % in Belgium. As illustrated in the graph, certain socio-demographic factors in some countries contribute negatively to the concentration index. This is particularly evident for household size (notably in Belgium and Spain) and socio-economic status (in Slovakia and Denmark). These factors mitigate the influence of the income variable, leading to a more uniform distribution of energy expenditures along the income distribution. This effect can also be explained as follows: while larger households tend to consume more energy, they may be more prevalent among lower-income groups, at least in some countries. The variable tenant status also leads to an increase in the concentration of energy expenditures. According to the data, 30.3 % of households in the lowest income quintile are tenants, compared to only 16.6 % in the highest quintile. Other variables, such as the number of income earners and socio-economic status, display mixed effects depending on the country. For instance, households with two income earners may either represent a

[18] The indicator variables for the different household type categories (1–6) capture the composition of the household, such as one-, two- or three-person household with or without children. The indicator variables for socio-economic status (1–6) represent manual worker and non manual workers in the private and public sector respectively, as well as self-employed or unemployed individuals. Additionally, we use indicator variables of the age of the household head (above 65, between 64 and 24 and below 24), the number of income earners (0–4, where 4 is for households with more than three income earners), and the household size (1–7, where 7 stands for all households with more than 6 household members).

[19] The step-by-step derivation is presented in Appendix B.

[20] For instance, for Bulgaria we multiply the coefficient for *Education2* ($\beta_{kjt} = 123.75$) with the mean of *Education2* ($\bar{x}_{kjt} = 0.15$) and divide it by mean energy expenditures in Bulgaria in 2020 ($\mu_{jt} = 1095.35$) to get 0.02.

[21] For instance, to compute the contribution of household size to the concentration of energy expenditures, we multiply the concentration indices of the categories 'HHsize3', 'HHsize4', 'HHsize5', 'HHsize6' and 'HHsize7' by their respective elasticities and add the results. If we assume larger households spend more on energy (higher absolute value of β) and bigger households are more frequent among lower income categories (negative C_{kjt}), then household size will feature a negative contribution to the concentration of energy expenditures and thus smooth out the distribution of these along the income distribution (as for instance, in Spain). If for instance, the opposite holds and bigger households are rather present among richer house-holds, this will increase the concentration of energy expenditures (for instance, in the illustrative example in Bulgaria).

[22] We capture tenant status by a binary indicator illustrating whether a household reports positive rental expenditures or not. However, using this as a proxy for tenant status must be treated with caution, since some households may not pay rent due to arrangements with family members or because rents are covered by the government. Even though this variable does not perfectly capture tenant status, the resulting shares of tenants in the overall population by country are largely consistent with official statistics. The estimated contributions of tenant status to the concentration index vary between 0 % in Slovakia and 27 % for Belgium.

[23] The calculated concentration indices slightly differ from those in Table 2 because observations with missing values for the socioeconomic variables required for the decomposition analysis had to be excluded.

Table 3
Coefficients, elasticity and the corresponding concentration indices for Bulgaria and Spain 2020.

Country Variable	Bulgaria				Spain			
	Coefficient	SE	Elasticity	CI	Coefficient	SE	Elasticity	CI
Education2	123.75	72.23	0.02	−0.45	136.50*	56.06	0.01	−0.24
Education3	139.90*	69.05	0.07	−0.04	149.42*	58.47	0.01	−0.13
Education6	154.63*	72.22	0.04	0.36	234.83***	57.12	0.05	0.22
Equivalent income	0.10***	0.01	0.37	0.28	0.04***	0.00	0.25	0.29
Age HH head >64	236.67	259.61	0.10	0.19	377.14	374.19	0.10	0.01
Age HH head >65	295.56	261.35	0.15	−0.15	265.35	376.62	0.04	−0.02
HH type 2	383.61***	32.19	0.13	0.08	852.07***	44.51	0.10	0.10
HH type 3	484.62***	176.25	0.04	0.23	1283.62***	154.72	0.06	0.01
HH type 4	335.98***	107.20	0.01	−0.05	651.90***	99.69	0.01	−0.39
HH type 5	523.97***	172.85	0.06	0.21	1177.45***	149.69	0.13	−0.02
HH type 6	553.19***	189.65	0.03	−0.04	1263.22***	167.91	0.03	−0.19
HH size 3	248.17	165.05	0.03	0.23	275.91*	138.74	0.02	0.00
HH size 4	461.22***	170.43	0.04	0.20	764.88***	145.32	0.06	−0.05
HH size 5	591.32***	188.33	0.01	−0.08	1075.06***	159.65	0.02	−0.19
HH size 6	1071.06***	216.32	0.01	−0.29	998.24***	210.74	0.00	−0.38
HH size 7	610.14***	236.06	0.00	−0.37	1404.61***	266.87	0.00	−0.62
1 income earner	28.99	44.72	0.01	0.12	−16.19	61.16	−0.00	−0.06
2 income earners	244.05***	66.23	0.06	0.31	131.08	78.82	0.02	0.10
3 income earners	405.77***	103.17	0.02	0.26	259.78*	117.31	0.01	−0.01
4 income earners	438.40***	135.50	0.01	−0.09	377.51*	167.09	0.00	−0.02
Socio.economic status 2	38.17	45.69	0.00	0.57	−50.29	43.36	−0.00	0.35
Socio.economic status 3	179.26***	57.01	0.01	0.26	273.22***	50.61	0.01	−0.00
Socio.economic status 4	−239.64***	56.66	−0.01	−0.43	−446.27***	68.18	−0.01	−0.67
Socio.economic status 5	4.87	51.93	0.00	−0.20	−34.20	67.85	−0.00	−0.00
Socio.economic status 6	−52.59	74.52	−0.00	−0.25	−275.61***	78.55	−0.01	−0.33
Tenant	−188.62*	83.45	−0.00	0.34	−374.16***	39.06	−0.02	−0.14

Notes: 1 % significance level (**), 5 % significance level (**). We use 2938 observations from Bulgaria and 18,379 from Spain in the year 2020. The dummy variables "socio-economic status" and "HH type" indicate the socio-economic and demographic composition of the household. The following categories apply: one adult (1), two adults (2), more than two adults (3), one adult with children (4), two adults with children (5), more than two adults with children (6). Manual worker (1), non-manual worker (2), self-employed worker or farmer (3), unemployed (4), retired (5), other inactive (6). The reference category is manual worker and omitted from the present listing. The reference category for all indicators is represented by the lowest value of the indicator. For example households with no income earner is the reference category. The variable household size 1 does not appear in the table as it is multicollinear to HH type 1 and hence household size 2 is the reference category.

Table 4
Concentration indices, cumulative contribution of explanatory variables and explained fraction Cumulative Contribution.

Country	CI	$\sum_k \frac{\beta_k \overline{x_k}}{\mu} C_k$	Fraction (in %)	Residual
AT	0.09	0.09	96	0.00
BE	0.03	0.03	97	0.00
BG	0.21	0.18	85	0.03
CY	0.15	0.11	76	0.04
DE	0.13	0.11	88	0.02
DK	0.10	0.10	101	0.00
EE	0.12	0.11	90	0.01
EL	0.12	0.11	88	0.01
ES	0.09	0.08	93	0.01
FR	0.15	0.12	83	0.02
HR	0.17	0.16	95	0.01
HU	0.17	0.15	85	0.03
LT	0.15	0.13	89	0.02
LV	0.22	0.18	82	0.04
SK	0.09	0.08	90	0.01

Source: Own calculations based on data from Eurostat. Malta, Slovenia, Luxembourg, and the Netherlands were excluded from the analysis because data for the socioeconomic variables required for the decomposition were missing.

significant portion of high-income households—where both adults work due to the high opportunity costs of staying home—or a large share of low-income households that require dual incomes to cover basic needs. The age of the household head can, in some cases, reduce energy expenditure concentration. Retired individuals, who are often lower on the income distribution, tend to spend more time at home, leading to higher energy consumption.

5. Policy implications and conclusion

5.1. Policy implications

The above analysis revealed that the simple comparison of energy expenditure shares for different income categories is not sufficient for an accurate assessment of the magnitude of the decline in energy expenditure ratios along the income distribution. Using this first approach, countries such as Croatia, Greece or Latvia would appear to have the largest discrepancy between the expenditure shares of the lowest compared to the highest incomes. In contrast, employing the degree of convexity (DOC) or the Kakwani index delivers a different ranking. Using the DOC, we can detect whether a smaller subgroup is disproportionately affected—something that may be obscured in an aggregated quintile-based analysis. It informs us how *sharply* energy burdens fall as income rises — not just that the poor pay more, but how much more, and how quickly the burden eases. A steeper (more convex) decline implies severe burdens at the bottom, highlighting the need for targeted relief for the lowest-income groups. It helps calibrate threshold-based interventions like for instance, energy vouchers. This is the case for countries such as the Netherlands. Finally, the Kakwani index allows for an aggregated assessment of the effect of energy expenditures within a country. Its function is comparable to other widely recognized inequality measures in the public economics literature, such as the Gini or Atkinson index, enabling policymakers to systematically compare between countries, regions and years and evaluate policy reforms (e.g., carbon pricing) based on their distributional impacts. Between 2010 and 2020, the burden of energy expenditures as measured by the Kakwani index increased in nearly all countries, with particularly strong increases observed in Lithuania and Estonia.

To translate our findings into actionable policy, several strategic considerations must be addressed—particularly the need to balance

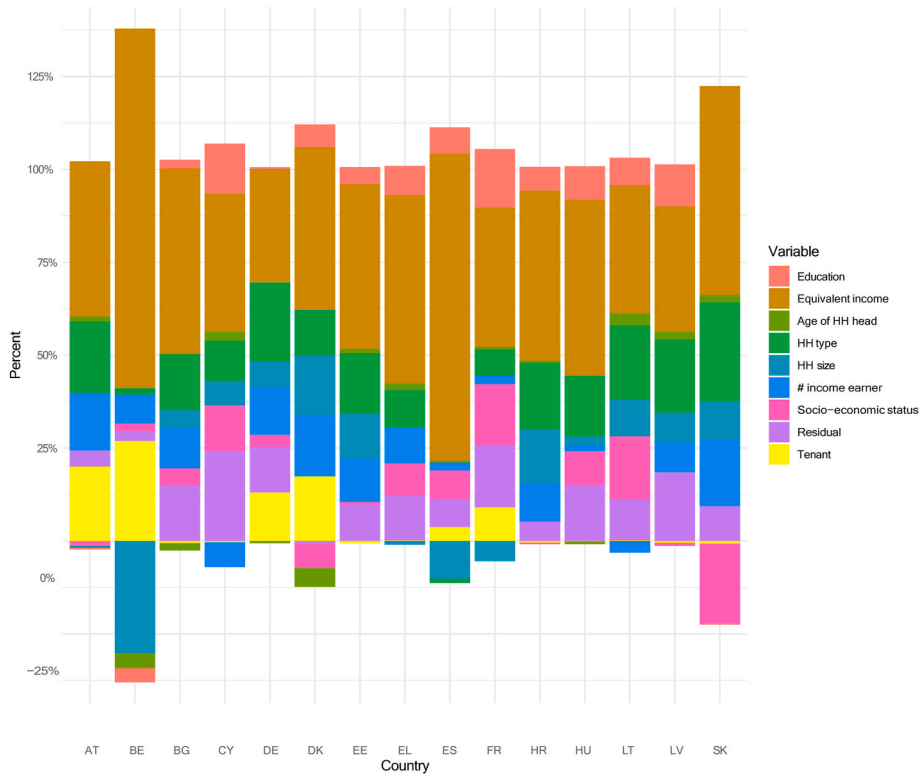

Fig. 7. Relative contribution of different socio-demographic variables to explaining the concentration index for various European Union states in 2020.
Source: Own illustration based on data from Eurostat. Malta, Slovenia, Luxembourg, and the Netherlands were excluded from the analysis because data for the socioeconomic variables required for the decomposition were missing.

targeted economic relief with broader climate objectives.

First, targeted income support or energy vouchers may be suitable when energy costs disproportionately impact a small segment of low-income households without heavily burdening the majority. These can provide effective relief without unduly straining state finances. Evaluating the degree of convexity in energy expenditure shares offers insight into whether a narrowly defined low-income group or a broader segment, including middle-income households, is adversely affected. The comparison between Austria and the Netherlands that feature similar Kakwani indices but different DOCs, illustrates that in the Netherlands, there is a large discrepancy between the share of energy expenditures in income for households at the lowest end of the income distribution compared to the rest, whereas this discrepancy is less pronounced in Austria.

Second, examining the underlying causes of high energy expenditures among specific socio-economic groups can provide insights into the root of the issue and inform targeted solutions. An analysis of the heating types of affected households is crucial to find long-run solutions. For example, if low-income households primarily depend on fossil-based heating, subsidies to facilitate a transition to renewable heating systems could prove effective. This is the case for Croatia where households using coal-based heating systems bear the highest energy cost burden relative to their income. Supporting the shift to renewable energy not only mitigates these impacts, but also reduces reliance on costly fossil fuels. Policies that promote affordable access to renewable energy and enhance energy efficiency would alleviate financial burdens on vulnerable households while simultaneously advancing environmental objectives. A further aspect to consider relates to the share of tenants within the affected population. Energy efficiency investments are undertaken by the landlords, not the tenants. Since the former bear the costs and the latter enjoy the benefits, investments into energy efficiency may fall short of the optimal amount. This leads to higher housing related energy expenditures among tenants compared to homeowners. This is then especially problematic in countries with a high share of tenants in the overall population.

Third, when energy costs impact not just low-income households but a broader segment of the population, improving energy efficiency or modifying the household energy mix may be more effective than direct transfer payments. This is exemplified by Bulgaria, which exhibits both a high DOC and a consistently high share of income spent on energy across all income quintiles. While small disparities in energy spending across income groups may seem manageable, high overall energy costs can pose significant challenges. Income-based measures, such as credits or vouchers, may strain public budgets without addressing the root causes of the issue. In such scenarios, policies aimed at enhancing energy efficiency or transitioning the energy mix toward renewables can provide a more sustainable solution.

Fourth, addressing income inequality and incorporating broader socio-demographic factors is essential. Reducing income inequality can contribute to a more equitable distribution of energy expenditures and help mitigate the effects of rising energy costs. However, our de composition analysis reveals that factors beyond income—such as household size or tenant status—also significantly influence energy expenditure patterns. For instance, in Belgium, household size reduces the concentration of energy expenditures, likely due to a higher prevalence of larger households in lower income brackets. As larger households tend to consume more energy overall, this leads to a more even distribution of energy spending. Consequently, policy interventions should not rely solely on income-based transfers. Even within the same income group, households may experience the burden of energy costs differently—for ex-ample, depending on household size—highlighting the need for more nuanced and targeted support measures.

An important limitation of our methods is that households with very low incomes and simultaneously a small share of energy expenditures relative to income may in fact suffer from hidden energy poverty. Our aggregated measures are likely to underestimate the magnitude of these

effects in such cases, since the overall energy expenditures of low-income households appear lower, the corresponding budget share is reduced, and the decline in the energy share across the income distribution is thus flattened. In terms of Lorenz curves, this phenomenon translates into an increase in convexity; however, if only a few households are affected, the effect may remain less visible. Still, this is an important issue for economic policy, as these households may lack sufficient means to cover essential energy needs such as heating or cooking. Hidden energy poverty of this kind can be identified when low energy expenditures coincide with very low incomes, whereas low expenditures at high income levels may also reflect self-production (e.g. via solar panels). Targeted measures such as energy credits or vouchers may be the most effective instruments to relieve this kind of energy poverty.

Climate change mitigation policies, particularly those involving higher carbon prices, may increase the share of household income spent on energy. Lorenz curves provide a useful tool for examining the distributional effects of taxing energy goods. The observed decline in the concentration of energy expenditures across most European countries between 2010 and 2020 provide initial evidence of a potentially regressive impact of such policies. How-ever, these developments are not driven by carbon pricing alone. Broader elements of the energy transition—such as the uptake of solar panels and electric vehicles—also influence the distribution of energy expenditures, as these technologies enable some households to reduce their energy costs. The existing literature mostly focuses on the distributional implications of government subsidies for climate-friendly technologies, such as solar panels (Lekavičius et al., 2020), electric vehicles (Guo and Kontou, 2021), or weatherization (Borenstein and Davis, 2024). These subsidies often predominantly benefit higher-income households. Since these households are more likely to adopt such technologies, this trend is likely to be increasingly reflected in the future distribution of energy expenditures. Households with electric vehicles and/or solar panels typically face lower energy costs, all else being equal. Accordingly, more recent expenditure data may already reflect this shift. Future research should therefore ex-amine how changes in energy production and technology adoption affect inequality, going beyond the impact of carbon pricing alone.

Any policy addressing climate change mitigation must be carefully adapted to the specific economic and social conditions prevailing in each country. While some fiscal instruments may conflict with climate goals, others can advance both economic equity and environmental sustainability. Notably, certain energy price increases are intentional, aimed at incentivizing environmentally friendly behavior. Thus, addressing the regressive effects of such policies necessitates complementary strategies that enhance equity without undermining incentives for sustainable, low-carbon choices.

5.2. Conclusion

This paper contributes to the body of research on energy expenditure inequality in advanced economies by quantifying the magnitude of the differences in household energy expenditure ratios along the income distribution. In the context of the targeted reductions in greenhouse gas emissions and the necessary transformation of energy systems, this consideration is of paramount importance. Policy instruments such as CO_2 prices or subsidies aimed at enhancing energy efficiency in buildings, promoting the adoption of solar panels, and supporting the expansion of electromobility have a major impact on household energy expenditures. At the same time, households considerably differ in terms of socio-demographic characteristics, budget constraints or living conditions (e.g., tenants versus homeowners). Thus, these instruments and hence energy prices do not have a uniform effect in the population. We analyze cross-sectional survey data from European countries for the years 2010, 2015, and 2020, revealing a consistent pattern of declining energy expenditure shares with higher incomes across all countries and years examined. However, there is significant variability among the studied countries.

A key aspect of our analysis is the argument that assessing inequality in the distribution of energy expenditure burdens among households requires a more nuanced approach than simply focusing on the proportion of income spent on energy. Our findings demonstrate that the countries with the lowest or highest share of equivalent disposable income spent on energy — or those with the largest absolute difference between the shares of income spent on energy by households in the lowest and highest income quintiles — do not necessarily correspond to those with the lowest or highest Kakwani index.

The design of effective climate and energy policies requires consideration of multiple dimensions to ensure that heavily burdened households are not disproportionately strained, thereby fostering political support for climate initiatives. The Kakwani index which has been applied in public economics to measure the progressivity of tax systems can be applied in this context as well. Furthermore, the degree of convexity in energy expenditure shares reveals whether a specific segment of low-income households is disproportionately affected, or if the impact extends to a broader group, including the middle class. Policy instruments should be tailored based on these insights.

The focus on Europe does not constrain the generalization of our results. Many industrialized countries have already implemented stringent climate regulations, and their experiences offer valuable lessons for other regions. Beyond the EU, countries such as China, with its commitment to net-zero targets by 2060 provide relevant contexts for comparison.

Interestingly, we find that socio-demographic factors, such as household size or socio-economic status, have a negative contribution to the concentration index in certain countries, thereby contributing to a more uniform distribution of energy expenditures across the population. This finding is significant because it underscores that distinct policies addressing the unequal distribution of the energy expenditure burden among households may be warranted beyond income-based transfers. Further research should delve deeper into these channels. While this study primarily examines the distributional effects of energy expenditures, we still believe that it can enable a more informed assessment of the impact of rising energy price, insofar as they are reflected in higher energy expenditures.

CRediT authorship contribution statement

Ivan Ackermann: Writing – original draft, Methodology, Data curation. **Doina Radulescu:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing pro-cess

During the preparation of this work the authors used CHATGPT mainly for the introduction, policy implications and conclusion in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ivan Ackermann reports financial support was provided by Swiss Federal Office of Energy. Doina Radulescu reports a relationship with Energie Wasser Bern that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

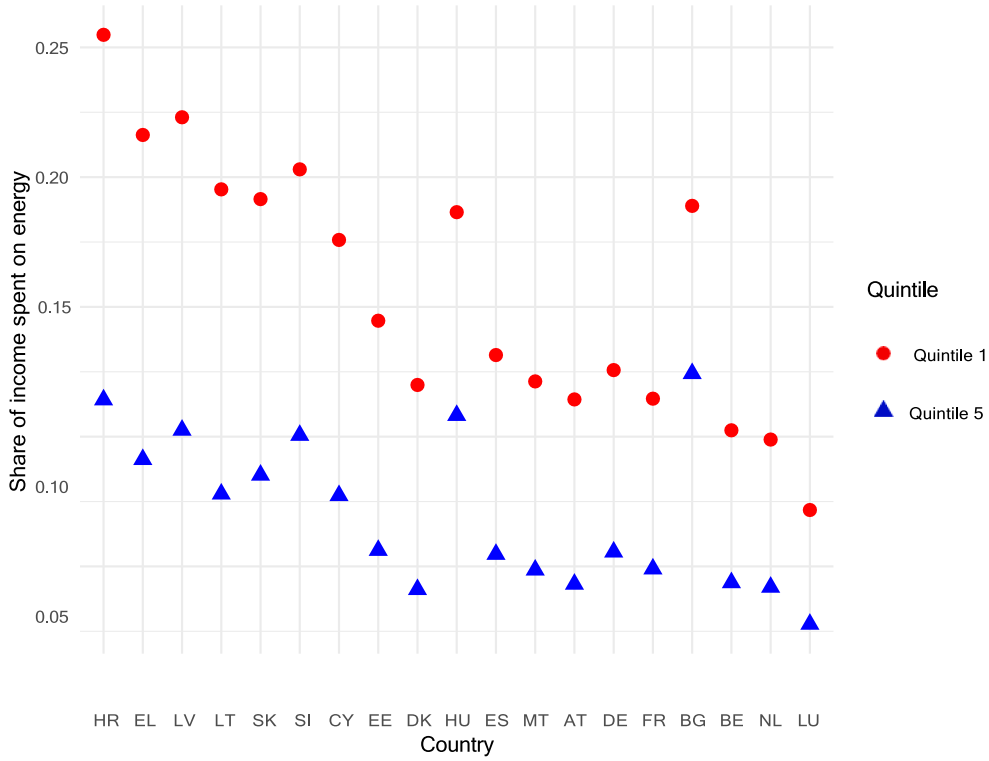

Fig. A1. Shares of total energy expenditures in equivalent income for the first and fifth income quintile in 2020
Source: Own illustration based on data from Eurostat. Includes only households having a share of income spent on energy between 0 and 50 %.

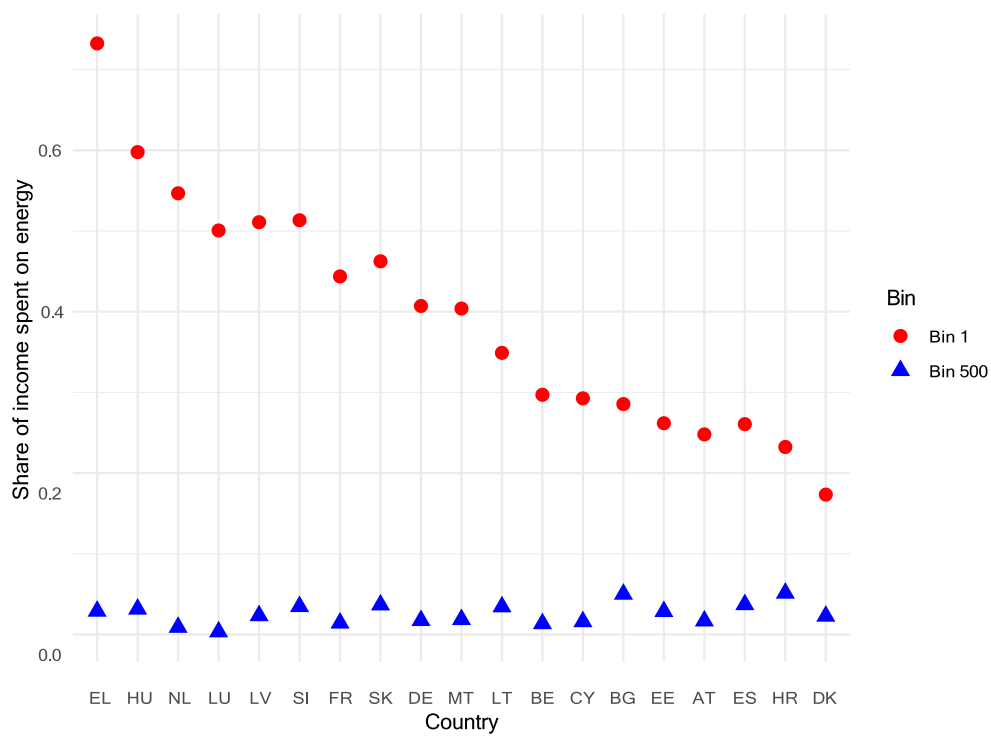

Fig. A2. Shares of total energy expenditures in equivalent income for the first and 500th income bin in 2020
Source: Own illustration based on data from Eurostat. The graphic shows the first bin (lowest income group) and the 500th for each country in 2020. They are ordered from highest difference to lowest from left to right.

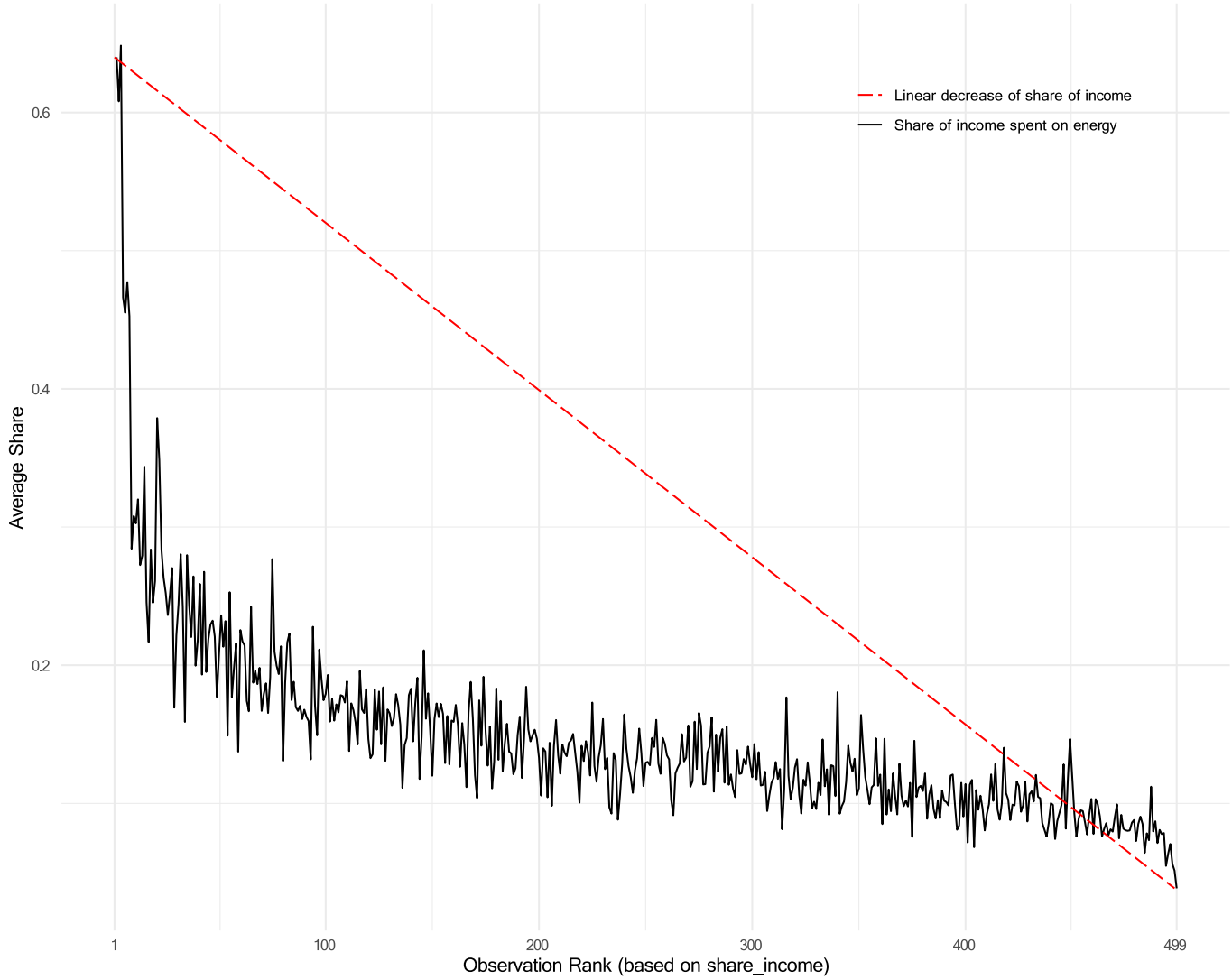

Fig. A3. Actual vs. linearly decreasing shares of energy expenditures in income for Greece in 2020 - excluding the 1st and 500th income bin
Source: Own illustration based on data from Eurostat.

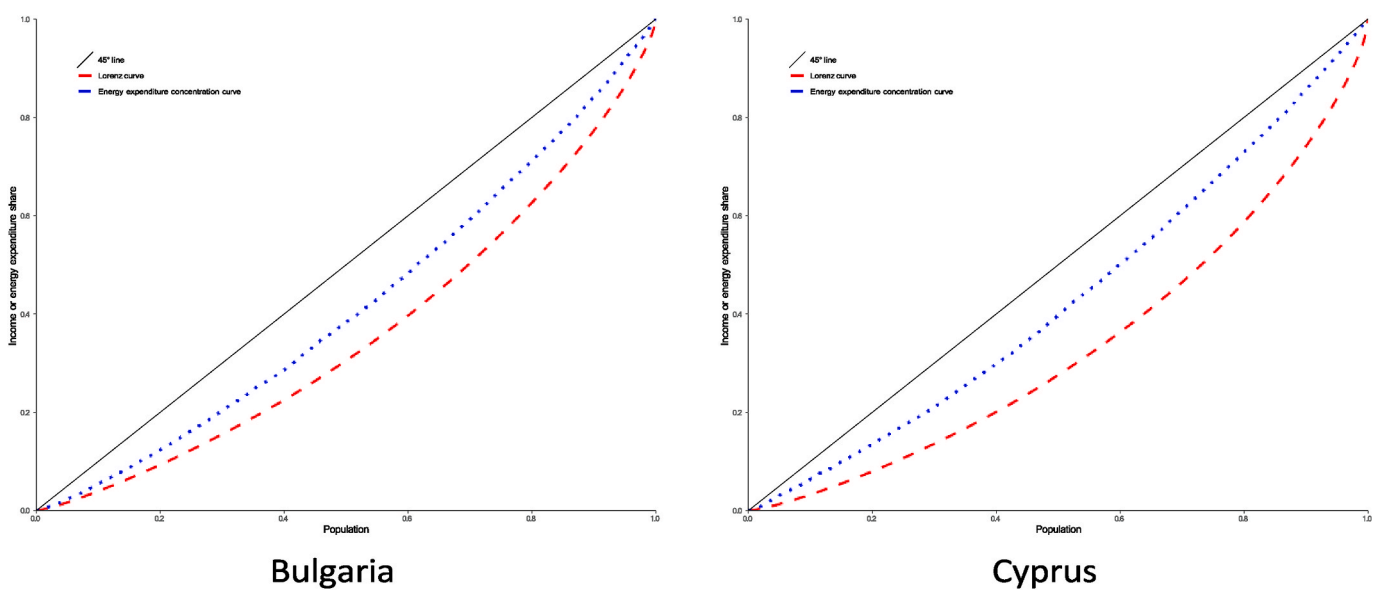

Fig. A4. Comparision of Lorenz and concentration curves for Bulgaria and Cyprus in 2020
Source: Own illustration based on data from Eurostat.

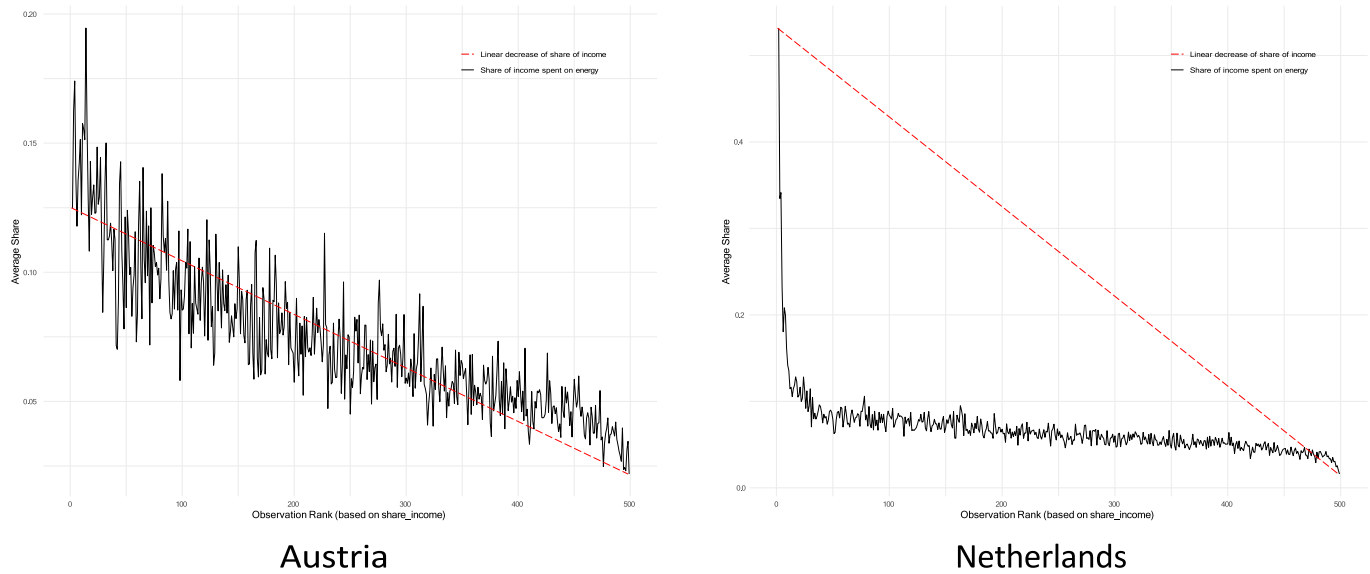

Fig. A5. Comparison of DOC for Austria and the Netherlands in 2020
Source: Own illustration based on data from Eurostat.

Table A1
Difference in heating system between Q1 and Q5 and average share of income spent on energy by heating system

Country	△ Share Coal	△ Share Gas	△ Share Heat	△ Share Oil	△ Share Electric	Coal	Electric	Gas	Heat	Oil
AT	0.007	−0.053	0.083	−0.044	0.007	0.086	0.056	0.070	0.070	0.098
BE	0.006	−0.033	0.000	0.005	0.022	0.084	0.059	0.065		0.091
BG	0.150	−0.028	−0.184	−0.002	0.063	0.220	0.129	0.145	0.145	0.104
CY	−0.053	0.401	0.000	−0.259	−0.089	0.132	0.122	0.125		0.110
DE	−0.011	−0.105	0.118	0.029	−0.030	0.106	0.069	0.090	0.087	0.130
DK	−0.012	−0.135	0.195	−0.009	−0.040	0.084	0.050	0.051	0.080	0.078
EE	0.163	−0.027	−0.079	0.000	−0.057	0.131	0.071	0.081	0.098	0.109
EL	0.136	−0.046	−0.008	−0.206	0.124	0.167	0.118	0.127	0.155	0.151
ES	0.011	−0.248	0.000	−0.033	0.271	0.112	0.084	0.089		0.110
FR	−0.057	0.193	−0.001	−0.034	−0.101	0.086	0.069	0.088	0.075	0.108
HR	0.476	−0.362	−0.102	−0.025	0.013	0.234	0.098	0.156	0.120	0.184
HU	0.024	0.009	−0.011	0.000	−0.022	0.182	0.084	0.142	0.111	
LT	0.248	−0.127	−0.098	0.001	−0.024	0.157	0.107	0.118	0.141	0.214
LU	−0.026	−0.111	0.000	0.077	0.060	0.044	0.037	0.045		0.068
LV	0.263	−0.073	−0.158	−0.010	−0.022	0.165	0.111	0.140	0.150	0.154
MT	0.001	0.075	0.001	−0.001	−0.076	0.095	0.074	0.102	0.096	0.202
NL	0.000	0.000	0.000	0.000	0.000			0.068		
SI	0.168	−0.119	−0.002	−0.097	0.051	0.157	0.128	0.130	0.120	0.166
SK	0.201	−0.162	−0.019	−0.003	−0.018	0.168	0.089	0.164	0.121	0.169

Note: The first six columns present the difference in the share of households using a specific heating system between the first and fifth income quintiles. The last five columns depict the average share of income allocated to energy expenditures by heating system.

Table A2
Degree of convexity (100 bins)

Country	2010	2015	2020
AT			−0.024
BE	−0.084	−0.040	−0.025
BG	−0.048	−0.011	0.028
CY	−0.037	−0.020	−0.023
CZ	−0.021	0.018	
DE	−0.024	−0.025	−0.044
DK	−0.047	−0.021	−0.045
EE	−0.054	−0.071	−0.061
EL	−0.070	−0.141	−0.188
ES	−0.066	−0.057	−0.033
FI	−0.002	−0.040	
FR	−0.079	−0.092	−0.092
HR	−0.066	−0.053	−0.077
HU	−0.062	−0.092	−0.098
IE	−0.121	−0.078	
LT	−0.035	−0.039	−0.049

(*continued on next page*)

Table A2 (*continued*)

Country	2010	2015	2020
LU		−0.041	−0.047
LV	−0.091	−0.054	−0.095
MT	−0.094	−0.063	−0.063
NL		−0.131	−0.143
PL	−0.036	−0.027	
PT	−0.045	−0.106	
RO	0.013	0.006	
SE	−0.030	−0.031	
SI	−0.162	−0.138	−0.081
SK	0.026	−0.038	−0.061
UK	−0.078		

Note: This table presents the degree of convexity of energy expenditure shares along the income distribution for the years 2010, 2015, and 2020, based on the methodology described in section 3.3.

Table A3
Overview of changes in the inequality of energy expenditures in 16 European Union countries from 2010 to 2020.

Country	ΔCI	ΔKI
BE	−0.025	0.017
BG	−0.031	−0.049
CY	−0.019	−0.028
DE	−0.022	−0.013
DK	−0.043	−0.030
EE	−0.066	−0.070
EL	−0.066	−0.030
ES	−0.016	−0.007
FR	−0.027	−0.014
HR	−0.030	−0.024
HU	0.029	0.029
LT	−0.090	−0.085
LV	−0.006	−0.014
MT	−0.027	−0.037
SI	0.006	0.013
SK	−0.058	−0.044

Notes: Austria, Luxembourg, and the Netherlands are not included in the graph, as data are missing for either 2010 or 2020, making a comparison infeasible.

Table A4
Summary Statistics Kakwani Index

Kakwani Total	2010	2015	2020
Min	−0.22	−0.20	−0.21
Mean	−0.15	−0.15	−0.18
Median	−0.15	−0.16	−0.18
Max	−0.07	−0.08	−0.11
Kakwani Household	**2010**	**2015**	**2020**
Min	−0.29	−0.29	−0.28
Mean	−0.21	−0.21	−0.23
Median	−0.21	−0.20	−0.23
Max	−0.14	−0.12	−0.19
Kakwani Electricity	**2010**	**2015**	**2020**
Min	−0.29	−0.27	−0.25
Mean	−0.21	−0.22	−0.23
Median	−0.23	−0.23	−0.23
Max	−0.12	−0.17	−0.19
Kakwani Fuel	**2010**	**2015**	**2020**
Min	−0.15	−0.19	−0.19
Mean	−0.03	−0.04	−0.07
Median	−0.05	−0.07	−0.09
Max	0.25	0.24	0.16

Note: The distribution included 24 countries in 2010, 25 countries in 2015, and 19 countries in 2020, respectively.

Table A5
Summary Statistics Concentration Index

CI Total	2010	2015	2020
Min	0.08	0.05	0.04
Mean	0.14	0.13	0.11
Median	0.15	0.14	0.10
Max	0.23	0.23	0.17
CI Household	**2010**	**2015**	**2020**
Min	0.03	−0.00	−0.01
Mean	0.09	0.08	0.06
Median	0.08	0.08	0.05
Max	0.18	0.15	0.15
CI Electricity	**2010**	**2015**	**2020**
Min	0.01	−0.00	−0.01
Mean	0.08	0.07	0.06
Median	0.09	0.08	0.06
Max	0.17	0.15	0.15
CI Fuel	**2010**	**2015**	**2020**
Min	0.12	0.12	0.10
Mean	0.27	0.25	0.22
Median	0.24	0.21	0.19
Max	0.57	0.54	0.44

Table A6
Summary Statistics Gini Index

Gini	2010	2015	2020
Min	0.22	0.22	0.22
Mean	0.29	0.29	0.28
Median	0.29	0.28	0.28
Max	0.36	0.37	0.35

Table A7
Contribution of different socio-demographic variables to explaining the concen-tration index

Country Variable	Bulgaria		Spain	
	Contribution	%	Contribution	%
Concentration index	0.21	100	0.09	100
Education	0.01	2	0.01	7
Equivalent income	0.11	50	0.07	83
Age HH head	0.00	−2	0.00	0
HH type	0.03	15	0.00	−1
HH size	0.01	5	−0.01	−10
# income earner	0.02	11	0.00	2
Socio-economic status	0.01	5	0.01	8
Tenant	0.00	−1	0.00	3
Residual	0.03	15	0.01	7

Appendix B. Derivation of the Decomposition of the Concentration Index C_{jt}

The concentration index in country j and year t can be expressed as:

$$C_{jt} = \frac{2}{n \cdot \mu_{jt}} \sum_{i=1}^{n} exp_{ijt} R_{ijt} - 1 \tag{B.1}$$

where:

- exp_{ijt} denotes the expenditure by individual i in country j and year t,
- $\mu_{jt} = \frac{1}{n} \sum_{i=1}^{n} exp_{ijt}$ is the mean expenditure by country year,
- R_{ijt} is the fractional income rank of individual i in country j and year t.

Assume the following linear approximation for exp_{ijt} where x_{kijt} represent a number of control variables (i.e. income, age etc.) and β_{kjt} the corresponding coefficients:

$$exp_{ijt} = \sum_k \beta_{kjt} x_{ikt} + \varepsilon_{ijt} \tag{B.2}$$

Substituting eq. (B.2) into eq. (B.1) we get:

$$C_{jt} = \frac{2}{n\mu_{jt}} \sum_{i=1}^{n} \left(\sum_k \beta_{kjt} x_{ikt} + \varepsilon_{ijt} \right) R_{ijt} - 1 \tag{B.3}$$

$$= \frac{2}{n\mu_{jt}} \left(\sum_k \beta_{kjt} \sum_{i=1}^{n} x_{ikt} R_{ijt} + \sum_{i=1}^{n} \varepsilon_{ijt} R_{ijt} \right) - 1 \tag{B.4}$$

Let us define the concentration index of a regressor x_k similar to above as:

$$C_{kjt} = \frac{2}{n\overline{x_{kjt}}} \sum_{i=1}^{n} x_{ikt} R_{ijt} - 1 \tag{B.5}$$

Rearrange:

$$\sum_{i=1}^{n} x_{ikt} R_{ijt} = \frac{n\overline{x_{kjt}}}{2} \left(C_{kjt} + 1 \right) \tag{B.6}$$

Ignoring constants and focusing on deviations (as the −1 cancels in aggregate), we get the approximation:

$$\sum_{i=1}^{n} x_{ikt} R_{ijt} \approx \frac{n\overline{x_{kjt}} C_{kjt}}{2} \tag{B.7}$$

Substitute eq. (B.7) into eq. (B.4) we get:

$$C_{jt} = \sum_k \frac{\beta_{kjt} \overline{x_{kjt}}}{\mu_{jt}} C_{kjt} + \frac{GC_{\varepsilon jt}}{\mu_{jt}} \tag{B.8}$$

where $GC_{\varepsilon jt}$ denotes the generalized concentration index of the residuals:

$$GC_{\varepsilon jt} = \frac{2}{n} \sum_{i=1}^{n} \varepsilon_{ijt} R_{ijt} \tag{B.9}$$

Data availability

The authors do not have permission to share data.

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